



## CONTACT

EMAIL  
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WEB  
hsandmann.github.io

GITHUB  
hsandmann

LINKEDIN  
in/hsandmann

PROJECT  
graphole.com

LOCATION  
São Paulo, Brazil

RELOCATION  
Open to relocation (on-site)

## SKILLS

### Languages

Java Python Go Kotlin C  
C++ JavaScript SQL

### Machine Learning & AI

Deep Learning Neural Networks  
Computer Vision NLP PCA SVM  
Clustering

### Backend & Architecture

Spring Boot Spring Cloud  
Microservices Design Patterns  
Software Architecture

### Data & Infrastructure

PostgreSQL Redis Kafka  
RabbitMQ Docker Kubernetes  
Spark MongoDB Neo4j

### Domains

Banking & Fintech  
Cryptography (PKI/HSM)  
High-Performance Computing  
POS/ERP

## EDUCATION

**D.Sc. • Electrical Engineering**  
Polytechnic School, USP • 2007–2012

**M.Sc. • Electrical Engineering**  
Polytechnic School, USP • 2003–2006

**B.Sc. • Computer Science**  
FEI • 1999–2002

## LANGUAGES

Portuguese Native  
English Advanced  
German B1

# Humberto Sandmann

Computer Scientist & Professor

## PROFILE

Senior software architect and artificial intelligence engineer with 20+ years building high-stakes systems across banking, fintech, and retail — core banking, cryptography (PKI/HSM), high-performance computing, computer vision, and cloud-native microservices. D.Sc. in neural computing; also a university professor teaching software design, AI, and distributed systems.

## EXPERIENCE

### Founder • H-Sandmann

2006–2025

Independent consultancy in machine learning and software architecture.

Machine Learning Software Architecture

### Co-Founder • SelsanTech

2012–2023

Co-founded a venture delivering machine-learning and software-architecture solutions.

Machine Learning Software Architecture

### Senior Software Architect & Machine Learning Engineer • RockSpoon

2018–2019

Architected POS/ERP/payroll platforms and computer-vision/ML features for a restaurant-tech company.

POS ERP JavaEE PostgreSQL OpenCV Computer Vision Go Android Kotlin

### Senior Consultant, Computational Optimization • Banco Votorantim

2015–2016

Return-on-equity optimization on high-performance computing infrastructure for a major bank.

RoE High-Performance Computing J2EE Spring Boot Jenkins Test Automation

### Senior Consultant, Computational Optimization • TecBan

2011

### Senior Software Engineer / Architect • ABN Amro Bank

2006–2007

### Senior System Analyst • Politec

2005–2006

### Senior Software Engineer / Architect • EverSystems

2004–2005

### Senior System Analyst • Central Online

2002–2003

### Senior System Analyst • Datamace

1999–2002

### Computer Programmer • Heraeus Electro-Nite

1998–1999

## ACADEMIC POSITIONS

### Associate Professor • Insper

2018–present

### Associate Professor • ESPM

2014–2025

### Associate Professor • FIAP

2012–2015

### Visiting Professor • Toledo Prudente Centro Universitário

2015

### Associate Professor • Universidade Cruzeiro do Sul

2012–2013

### Temporary Professor • Universidade Nove de Julho

2011

### Teaching Assistant • Polytechnic School, University of São Paulo

2008–2011

Full academic CV, publications and teaching record at [hsandmann.github.io](https://hsandmann.github.io)